

Chapter 7.3

Consigning operations concerning the packing and use of cargo transport units (CTUs) and related provisions

7.3.1 Introduction

This chapter contains the provisions appropriate to those responsible for the consignment operations in the dangerous goods transport supply chain, including provisions relating to packing of dangerous goods into cargo transport units.

7.3.2 General provisions for cargo transport units

7.3.2.1 Packages containing dangerous goods shall only be loaded in cargo transport units that are strong enough to withstand the shocks and loadings normally encountered during transport, having regard to the conditions to be expected during the anticipated journey. The cargo transport unit shall be constructed in such a way as to prevent the loss of contents. Where appropriate, the cargo transport unit shall be fitted with devices to facilitate securing and handling of the dangerous goods. The cargo transport units shall be adequately maintained.

7.3.2.2 Unless otherwise specified, the applicable provisions of the *International Convention for Safe Containers, 1972* (CSC Convention), as amended, shall be followed for the use of any cargo transport unit which meets the definition of a “container” within the terms of that Convention.

7.3.2.3 The *International Convention for Safe Containers, 1972* (CSC Convention) does not apply to offshore containers that are handled in open seas. The design and testing of offshore containers shall take into account the dynamic lifting and impact forces that may occur when a container is handled in open seas in adverse weather and sea conditions. The requirements for such containers shall be determined by the approving competent authority. Such provisions should be based on the *Guidelines for the approval of offshore containers handled in open seas* (MSC/Circ.860). Such containers shall be clearly marked with the words “OFFSHORE CONTAINER” on the safety approval plate.

7.3.3 Packing of cargo transport units

7.3.3.1 Prior to the use of a cargo transport unit it shall be checked to ensure that it is apparently fit for its intended purpose.

7.3.3.2 The interior and exterior of a cargo transport unit shall be inspected prior to loading to ensure that there is no damage that could affect its integrity or that of the packages to be loaded in it.

7.3.3.3 Packages shall be examined and any found to be damaged, leaking or sifting shall not be packed into a cargo transport unit. Care shall be taken to see that excessive water, snow, ice or foreign matter adhering to packages is removed before packing into a cargo transport unit. Whenever the handling provision “keep as dry as reasonably practicable” (H1) is assigned in column 16a of the Dangerous Goods List, the cargo transport unit including any contained goods, securing or packing materials shall be kept as dry as reasonably practicable.

7.3.3.4 Drums containing dangerous goods shall always be stowed in an upright position unless otherwise authorized by the competent authority.

7.3.3.5 Cargo transport units shall be loaded in accordance with 7.3.4, so that incompatible dangerous or other goods are segregated. Specific loading instructions such as orientation arrows, not to be double stacked, keep dry or temperature control requirements shall be met. Liquid dangerous goods shall be loaded below dry dangerous goods whenever possible.

- 7.3.3.6** Packages containing dangerous goods and unpackaged dangerous articles shall be secured by suitable means capable of restraining the goods (such as fastening straps, sliding slatboards, adjustable brackets) in the cargo transport unit in a manner that will prevent any movement during transport which would change the orientation of the packages or cause them to be damaged. When dangerous goods are transported with other goods (e.g. heavy machinery or crates), all goods shall be securely fixed or packed in the cargo transport units so as to prevent the release of dangerous goods. Movement of packages may also be prevented by filling any voids by the use of dunnage or by blocking and bracing. Where restraints such as banding or straps are used, these shall not be over-tightened to cause damage or deformation of the package or the securing points (such as D-rings) within the cargo transport unit. The packages shall be packed in such a way that there will be a minimum likelihood of damage to fittings during transport. Such fittings on packages shall be adequately protected. Where restraints such as banding or straps with integral container fittings are used, care should be taken to ensure that the Maximum Securing Load (*MSL*) of the fittings is not exceeded.
- 7.3.3.7** Packages shall not be stacked unless designed for that purpose. Where packages of different stacking designs are to be loaded together, consideration shall be given to their compatibility for stacking with each other. Where necessary, stacked packages shall be prevented from damaging the package below by the use of load bearing devices.
- 7.3.3.8** Cargo shall be entirely contained within the cargo transport unit without overhang or projections. Oversized machinery (such as tractors and vehicles) may overhang or project outside of the cargo transport unit provided that the dangerous goods integral to the machinery cannot leak or spill outside of the cargo transport unit.
- 7.3.3.9** During loading and unloading, packages containing dangerous goods shall be protected from being damaged. Particular attention shall be paid to the handling of packages during their preparation for transport, the type of cargo transport unit to be used for their carriage and to the method of loading or unloading, so that accidental damage is not caused through dragging or mishandling. Packages that appear to be leaking or damaged so that the contents may escape shall not be accepted for transport. If a package is found to be damaged so that the contents leak, the damaged package shall not be transported but moved to a safe place in accordance with instructions given by a competent authority or a designated responsible person who is familiar with the dangerous goods, the risks involved and the measures that should be taken in an emergency.
- Note 1:** Additional operational requirements for the transport of packagings and IBCs are provided in the special packing provisions for packagings and IBCs (see chapter 4.1).
- 7.3.3.10** When a dangerous goods consignment forms only part of the load of a cargo transport unit, it should, whenever possible, be packed adjacent to the doors with marks and labels visible, so as to be accessible in the event of an emergency or to facilitate inspection.
- 7.3.3.11** If the doors of a cargo transport unit are locked, the means of locking shall be such that, in cases of emergency, the doors can be opened without delay.
- 7.3.3.12** When venting is required, venting devices shall be kept clear and operable.
- 7.3.3.13** Cargo transport units containing dangerous goods shall be marked and placarded according to chapter 5.3. Irrelevant marks, labels, placards, orange panels, signs and marine pollutant marks shall be removed, masked or otherwise obliterated before packing a cargo transport unit.
- 7.3.3.14** Cargo transport units shall be packed so that the cargo is uniformly distributed.
- 7.3.3.15** If goods of class 1 are packed, the cargo transport unit shall comply with the definition in 7.1.2 for closed cargo transport unit for class 1.
- 7.3.3.16** If goods of class 7 are packed, the transport index and, if applicable, the criticality safety index, shall be limited according to 7.1.4.5.3.
- 7.3.3.17** Those responsible for the packing of dangerous goods into a cargo transport unit shall provide a "container/ vehicle packing certificate" (see 5.4.2). This document is not required for tanks.
- 7.3.3.18** Flexible bulk containers are not allowed to be transported in cargo transport units (see 4.3.4).

7.3.4 Segregation provisions within cargo transport units

- 7.3.4.1** Dangerous goods which have to be segregated from each other according to the provisions in chapter 7.2 shall not be transported in the same cargo transport unit with the exception of dangerous goods which shall be segregated "away from" each other which may be transported in the same cargo transport unit with the approval of the competent authority. In such cases an equivalent standard of safety shall be maintained.